
Progressive Music Context Understanding with Language Models, Music Graphs, Multimodal Fusion, and Contrastive Retrieval

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Abstract

Music context is expressed through both semantic descriptions and relations within the audio signal. We study this problem through four progressively broader tasks. First, a DistilBERT multi-label classifier maps MusicCaps captions to a training-derived context vocabulary and reaches a test Micro F1 score of 0.599 and mean Area Under the Precision Recall Curve of 0.502. Second, an audio-only GraphSAGE model represents FMA-small tracks as beat-synchronous segment graphs with temporal and acoustic-similarity edges; it improves Macro F1 score from 0.421 for a log-mel convolutional neural network to 0.481. Third, FMA-medium experiments combine GraphSAGE and DistilBERT using BERT-only, GNN-only, early-concatenation, and cross-attention ablations. Cross attention gives the highest genre accuracy at 0.761, while early concatenation gives the highest tag Mean Average Precision at 0.231. Finally, a MusicCaps GNN-BERT dual encoder trained with symmetric InfoNCE reaches Recall at 10 of 0.163 for caption-to-audio retrieval and 0.164 in the reverse direction, compared with 0.010 random chance. The study shows that semantic text, relational audio structure, supervised fusion, and contrastive alignment provide complementary forms of music-context understanding.

1 Introduction

Music understanding is inherently multimodal. A listener can describe genre, mood, instrumentation, production quality, and performance style in language, while the audio signal also contains repeated sections, temporal transitions, and related acoustic events. Text encoders such as BERT provide strong contextual semantic representations [Devlin et al., 2019, Sanh et al., 2019], while graph neural networks provide a natural way to represent relations among musical segments [Hamilton et al., 2017]. The central question in this project is how these two views of music complement each other.

We organize the study as a progression of capabilities rather than as four directly comparable benchmarks. Task 1 measures supervised semantic understanding from text only. Task 2 removes text and tests whether explicit music-structure graphs improve over a conventional spectrogram model. Task 3 combines structural audio and semantic text under supervised genre and tag objectives. Task 4 changes the objective from fixed-label prediction to cross-modal audio-text alignment and retrieval. Tasks 2 and 3 use the Free Music Archive (FMA) [Defferrard et al., 2017], while Tasks 1 and 4 use MusicCaps, released with MusicLM [Agostinelli et al., 2023]. Because datasets and objectives differ, the paper does not interpret the four task metrics as a single monotonic score.

The main contributions are threefold. First, we build leakage-aware segment graphs whose edges combine temporal adjacency with non-local acoustic similarity. Second, we conduct a controlled multimodal ablation across BERT-only, GNN-only, early-concatenation, and cross-attention models.

Table 1: Experimental progression across the four tasks. Each task adds a capability rather than extending one common benchmark.

Task	Dataset	Input	Objective	Test size
1	MusicCaps	Caption text	62-label context tagging	580
2	FMA small	Audio segment graph	8-way genre classification	800
3	FMA medium	Audio graph + metadata text	16-way genre + 48 tags	1,771
4	MusicCaps	Audio graph + caption	Bidirectional retrieval	969

Third, we extend the graph and language encoders to a contrastive dual-encoder retrieval setting and analyze both exact retrieval and shared-space geometry.

2 Related work

BERT introduced deeply bidirectional Transformer representations that can be fine-tuned for downstream language tasks [Devlin et al., 2019]. Task 1 uses DistilBERT, a compact distilled variant that preserves much of BERT’s language capability at lower computational cost [Sanh et al., 2019]. For relational audio modeling, we use GraphSAGE [Hamilton et al., 2017], which learns inductive neighborhood aggregation functions and therefore fits variable-size music graphs.

FMA provides open audio, metadata, tags, and official train, validation, and test splits for music information retrieval [Defferrard et al., 2017]. MusicCaps contains expert-written descriptions of 10-second music clips and was released with MusicLM [Agostinelli et al., 2023]. Recent music-language work has shown the value of joint audio-text spaces. MuLan learns a two-tower music-language representation for retrieval and zero-shot transfer [Huang et al., 2022]; MusCALL uses contrastive audio-language learning for music retrieval and zero-shot tagging [Manco et al., 2022]; and CLAP extends contrastive language-audio pretraining to general audio concepts [Elizalde et al., 2023]. Task 4 follows this dual-encoder contrastive principle but replaces a standard audio tower with a GraphSAGE encoder over music structure graphs.

3 Experimental progression

3.1 Datasets, splits, and targets

Table 1 summarizes the four experiments. Task 1 uses all 5,521 MusicCaps metadata rows and creates a project-specific stratified split before selecting 62 target aspect phrases from training data only. Task 2 uses FMA-small and preserves the official artist-disjoint split after seven audio or graph construction failures. Task 3 uses FMA-medium with 16 top-level genres and 48 training-selected tag targets; 16 tracks fail robust audio or graph preprocessing. Task 4 uses the official MusicCaps metadata for provenance and an audio-enabled mirror containing 5,352 available pairs. The balanced MusicCaps subset supplies 969 audio-available test pairs.

For FMA experiments, the official artist-disjoint split is retained. In all graph experiments, feature normalization is fitted on training nodes only. Target vocabularies for Tasks 1, 3, and 4 zero-shot analysis are selected from training annotations only. These choices prevent preprocessing or vocabulary selection from using test information.

3.2 Audio graphs

A track is segmented using detected beats, with fixed-duration fallbacks when beat tracking produces too few or too many segments. Consecutive segments are connected bidirectionally, and non-adjacent segments are additionally connected when cosine similarity exceeds 0.50, with at most three added neighbors per node. Task 2 uses 228-dimensional node features including log-mel statistics and handcrafted descriptors. Tasks 3 and 4 use compact 100-dimensional descriptors based on MFCC, chroma, spectral contrast, tonnetz, and spectral statistics.

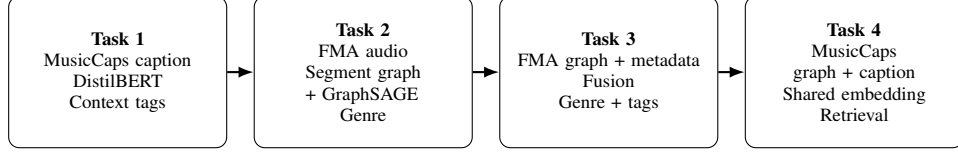


Figure 1: Progression of the study. Task 1 establishes supervised language semantics, Task 2 adds explicit audio structure, Task 3 combines both modalities, and Task 4 replaces fixed-label prediction as the primary objective with cross-modal retrieval. Arrows indicate increasing capability, not parameter transfer between tasks.

The graph encoder uses two residual GraphSAGE layers followed by global mean pooling. For a node i at layer ℓ , the update can be written compactly as

$$h_i^{(\ell+1)} = \text{LN}\left(h_i^{(\ell)} + \phi\left(\text{SAGE}\left(h_i^{(\ell)}, \mathcal{N}(i)\right)\right)\right), \quad (1)$$

where ϕ contains GELU and dropout. This construction preserves a direct path for the original segment representation while aggregating relational context.

3.3 Text, fusion, and contrastive objectives

Task 1 fine-tunes DistilBERT for independent binary tag prediction. With text representation t and tag k ,

$$\hat{y}_k = \sigma(w_k^\top t + b_k), \quad (2)$$

and training uses Binary Cross Entropy with Logits.

Task 3 uses semantic FMA metadata text while excluding target track tags from the BERT input. The final two DistilBERT blocks are fine-tuned. Early concatenation joins projected graph and text vectors directly. Cross attention instead lets the graph representation query token-level text states:

$$A = \text{softmax}\left(\frac{(gW_Q)(H_{\text{text}}W_K)^\top}{\sqrt{d}}\right), \quad z = [g; A(H_{\text{text}}W_V)^\top]. \quad (3)$$

Genre prediction uses cross entropy and tags use weighted Binary Cross Entropy with Logits.

Task 4 uses BERT-base-uncased and the same residual GraphSAGE principle. Both towers are projected to 256 dimensions and L2 normalized. For a batch of paired graph and caption embeddings,

$$S_{ij} = \frac{\tilde{g}_i^\top \tilde{t}_j}{\tau}, \quad \mathcal{L}_{\text{NCE}} = \frac{1}{2} [\text{CE}(S, I) + \text{CE}(S^\top, I)], \quad (4)$$

where τ is learnable. This symmetric objective trains both retrieval directions.

4 Task 1: supervised semantic tagging

Task 1 retains every normalized MusicCaps aspect phrase with at least 100 positive training examples, producing 62 labels. The target matrix is sparse, with approximately 5% positive entries. DistilBERT is fully fine-tuned for 10 Epochs, and validation mean Area Under the Precision Recall Curve selects the final checkpoint. The best Epoch is 10, with training loss 0.108 and validation loss 0.116.

A fixed threshold of 0.5 is overly conservative. Validation calibration selects a global threshold of 0.27, increasing test Micro F1 score from 0.497 to 0.599 by improving recall from 0.347 to 0.559. Table 2 reports the main comparison. Per-label calibration raises Macro F1 score further, but it is treated as secondary because some validation labels have limited support.

The gap between Micro and Macro F1 is not only a threshold artifact. Per-label test F1 has a Pearson correlation of 0.592 with training support. Frequent concepts such as *low quality*, *noisy*, and *instrumental* perform strongly, while several lower-support concepts are rarely activated by a single threshold. MusicCaps also contains near-synonymous phrases such as *instrumental* and *instrumental music*, which fragments supervision. Task 1 therefore establishes a strong semantic baseline but contains no audio structure.

Table 2: Task 1 MusicCaps caption-to-tag results. The validation-selected global threshold is the primary operating point.

Setting	Micro F1	Macro F1	Precision	Recall	Mean AUPRC
Train prevalence	0.188	0.026	0.134	0.314	0.048
DistilBERT, 0.50	0.497	0.212	0.875	0.347	0.502
DistilBERT, 0.27	0.599	0.363	0.646	0.559	0.502
Per-label calibration	0.560	0.490	0.487	0.657	0.502

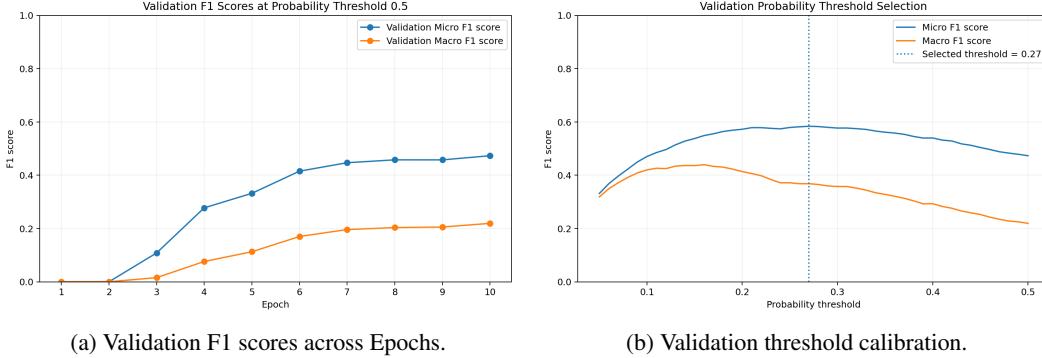


Figure 2: Task 1 diagnostics. Fine tuning steadily improves both F1 measures, but Macro F1 remains lower because of long-tail labels. Threshold calibration shows that 0.5 sacrifices recall and motivates the selected global threshold of 0.27.

5 Task 2: structural audio graphs

Task 2 tests whether relational structure improves audio-only genre recognition. FMA-small produces 7,993 valid cached graphs after seven construction failures. Graphs average 14.43 nodes, 26.85 directed temporal edges, and 58.80 directed similarity edges. The mean number of similarity edges per node is 4.06, confirming that the graph contains substantial non-local structure rather than only a temporal chain.

GraphSAGE is compared with a residual convolutional neural network operating on 128-bin log-mel spectrograms and with a majority predictor. The best GraphSAGE checkpoint occurs at Epoch 6. Training Macro F1 continues increasing afterward while validation performance becomes unstable, so early stopping is important.

GraphSAGE gains 6.25 percentage points in accuracy and 5.95 points in Macro F1 score over the convolutional neural network. Per-class F1 is strongest for Hip-Hop (0.725), Rock (0.638), International (0.602), and Electronic (0.570), while Folk (0.199) remains difficult. The 192-dimensional graph embeddings have a cosine silhouette score of only 0.036, indicating that audio structure alone does not form clean genre clusters. This motivates adding semantic information in Task 3.

6 Task 3: supervised multimodal fusion

Task 3 combines FMA-medium graph structure with semantic metadata text. Target track tags are excluded from BERT input text to prevent leakage. After robust preprocessing, 13,509 training, 1,704 validation, and 1,771 test pairs remain. The graph audit contains 16,984 valid graphs with 14.56 nodes and 59.61 similarity edges on average. The multi-label vocabulary contains 48 training-selected tags.

Table 4 shows that fusion behavior depends on the target. Cross attention gives the highest genre accuracy at 0.761, but BERT-only gives the highest genre Macro F1 score at 0.432. Early concatenation gives the strongest tag Mean Average Precision at 0.231 and the highest tag Macro F1 score at 0.161. GNN-only is weakest for semantic tags, with Mean Average Precision 0.046. Thus the audio and text branches are complementary, but cross attention does not dominate every metric.

Table 3: Task 2 test comparison on the balanced FMA-small test set. GraphSAGE improves all main metrics over the log-mel convolutional neural network.

Model	Accuracy	Macro F1	Micro F1	Macro AUPRC
Majority	0.1250	0.0278	0.1250	0.1250
Log-mel CNN	0.4213	0.4210	0.4213	0.4319
GraphSAGE	0.4838	0.4805	0.4838	0.4997

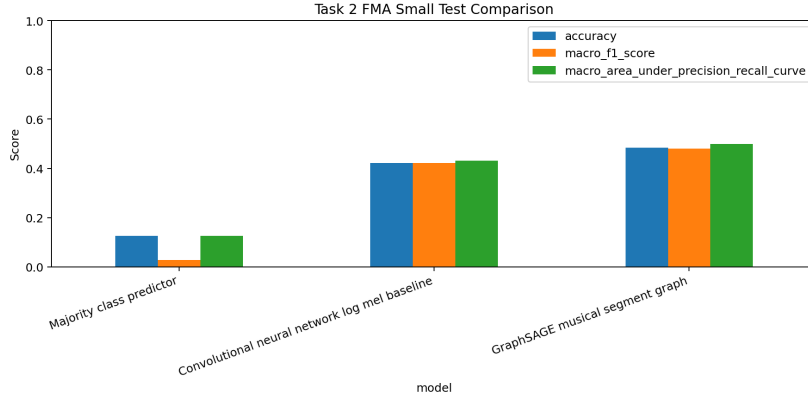


Figure 3: Task 2 model comparison. Explicit temporal and similarity relations improve accuracy, Macro F1 score, and Macro Area Under the Precision Recall Curve over the spectrogram baseline.

The cross-attention genre report reflects the heavy FMA-medium class imbalance. Electronic, Rock, Hip-Hop, Classical, Jazz, and Old-Time/Historic achieve strong F1 scores, while several small classes receive zero F1. This explains why the best accuracy and best Macro F1 occur in different ablations. Three repository case studies further connect graph structure and semantic attention. For example, track 13,814 is correctly classified as Folk with probability 0.954, and track 37,962 is correctly classified as Electronic with probability 0.964. A third Electronic example is classified correctly but misses its *downtempo* tag, illustrating that correct genre recognition does not guarantee fine-grained tag recovery. Because FMA does not provide aligned natural-language captions for every track, these case studies use semantic metadata rather than literal captions.

7 Task 4: cross-modal MusicCaps retrieval

Task 4 changes the primary objective from supervised labels to graph-caption alignment. The official MusicCaps release contains 5,521 metadata rows, while the audio-enabled mirror used for waveforms provides 5,352 unique available pairs. The project split contains 2,580 training, 1,803 validation, and 969 balanced-subset test pairs. All 5,352 graphs pass integrity checks. They average 9.81 nodes, 17.63 temporal edges, and 39.10 similarity edges, and 96.3% use beat-based segmentation.

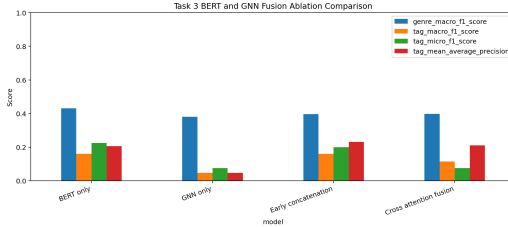
The dual encoder is selected by the mean validation Recall at 10 across both retrieval directions, with the best checkpoint at Epoch 11. Table 5 reports held-out retrieval. Caption-to-audio Recall at 10 is 0.163 and audio-to-caption Recall at 10 is 0.164, compared with 0.010 random chance.

The paired-to-random similarity gap is 0.300, but the modality centroids have cosine 0.693, leaving a modality gap of 0.307. Exact-pair retrieval is therefore meaningful but incomplete. This is also visible qualitatively: among ten stored caption queries, the true clip ranks range from 2 to 203. Non-paired high-ranked clips can still be musically plausible because many MusicCaps descriptions share instrumentation, production, and genre terms.

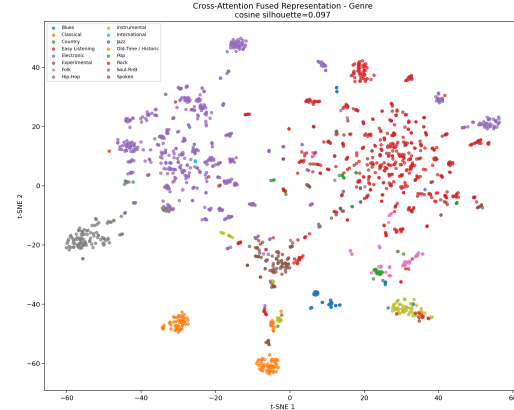
As a secondary transfer analysis, 40 training-selected MusicCaps aspect prompts are used for zero-shot prediction. Audio-graph prompting reaches Macro F1 score 0.117 and Macro Area Under the Precision Recall Curve 0.079, while caption prompting reaches 0.120 and 0.093. These weak scores show that instance-level alignment does not automatically provide a strong open-vocabulary tag classifier. We therefore treat retrieval as the principal Task 4 result.

Table 4: Task 3 FMA-medium ablation. Cross attention is strongest for genre accuracy, while early concatenation is strongest for tag ranking.

Model	G-Acc.	G-Macro F1	G-Micro F1	T-Macro F1	T-Micro F1	Tag mAP
BERT only	0.7386	0.4315	0.7386	0.1605	0.2249	0.2063
GNN only	0.6900	0.3814	0.6900	0.0479	0.0757	0.0464
Early concat.	0.7476	0.3963	0.7476	0.1612	0.2005	0.2313
Cross attention	0.7612	0.3980	0.7612	0.1155	0.0749	0.2103



(a) Four-model ablation.



(b) Cross-attention fused embeddings.

Figure 4: Task 3 analysis. The ablation exposes target-dependent fusion behavior. The fused representation shows local genre organization but substantial overlap, consistent with a cosine silhouette score of 0.097.

8 Discussion and limitations

The four tasks reveal a consistent division of labor between modalities. Task 1 shows that contextual text supervision is effective for explicit semantic concepts, but long-tail support and open-vocabulary synonym fragmentation limit Macro F1. Task 2 shows that explicit temporal and repeated-segment relations improve audio-only genre recognition over a spectrogram convolutional neural network. Task 3 shows that semantic metadata and graph structure are complementary, but also that the optimal fusion depends on whether the target is genre or tags. Task 4 shows that the same structural and language ideas can support retrieval without a fixed label vocabulary.

Several limitations prevent stronger cross-task claims. First, FMA and MusicCaps are different datasets, so Task 3 classification metrics and Task 4 retrieval metrics are not directly comparable. Second, Task 3 semantic input is FMA metadata rather than aligned captions or lyrics. Third, FMA-medium is highly imbalanced, so genre accuracy and Macro F1 can favor different models. Fourth, MusicCaps exact-pair retrieval counts only one paired clip as correct even when another retrieved clip is semantically compatible with the query. Finally, the MusicCaps waveform source contains 5,352 of the 5,521 official examples, so Task 4 evaluates the available subset rather than every official row.

9 Conclusion

This project develops music-context understanding from a text-only baseline to structural audio modeling, supervised multimodal fusion, and cross-modal retrieval. DistilBERT provides strong supervised semantics, GraphSAGE improves audio-only genre recognition over a log-mel convolutional neural network, fusion combines complementary structural and semantic cues, and contrastive learning creates a shared graph-text space that retrieves paired MusicCaps examples well above chance. The results suggest that music-context systems benefit from combining language and relational audio representations, while the best architecture and objective depend on whether the task requires fixed-label prediction or open-ended retrieval.

Table 5: Task 4 MusicCaps retrieval. Both directions substantially exceed random retrieval while exact rank-1 matching remains difficult.

Direction	R@1	R@5	R@10	MedR	MeanR	MRR
Caption to Audio	0.0330	0.1022	0.1631	69	137.5	0.0784
Audio to Caption	0.0237	0.0949	0.1641	69	129.4	0.0722
Random chance	0.0010	0.0052	0.0103	—	—	—

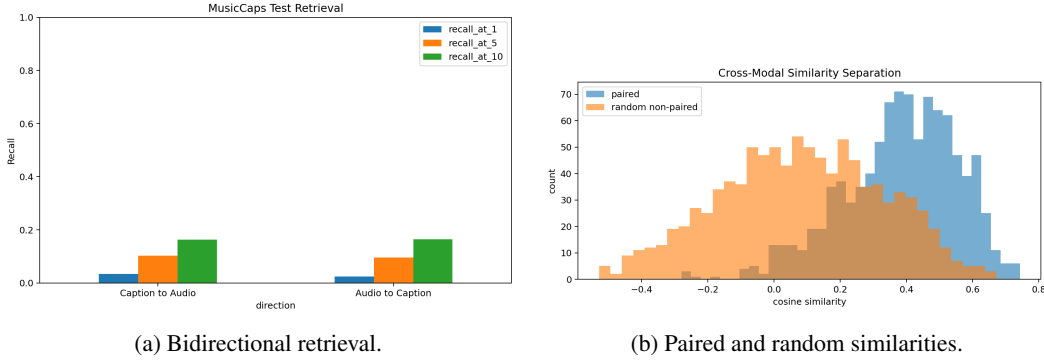


Figure 5: Task 4 test analysis. Recall increases consistently with retrieval depth, and the two directions are similar. True audio-caption pairs have mean cosine similarity 0.382 compared with 0.082 for random non-pairs, providing direct evidence that contrastive training learns a meaningful shared space.

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